THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET B

Chapters 1 & 2: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If one breath is about 0.5 litre and there are about 20 000 breaths a day, the air breathed in a day is about: 1

(a) 10 000 litres **(b)** 1000 litres **(c)** 1 lakh litres **(d)** 100 litres

2. Which of these is a good example of a scientific prediction? 1

(a) Saying the match will be lucky
(b) Using the speed and angle of a kick to find how far a football will travel
(c) Guessing the winner by colour of jersey
(d) Choosing numbers at random

3. The compass on the page numbers of the textbook stands for: 1

(a) memorising laws
(b) direction — choosing models and the right questions
(c) taking readings
(d) careful observation only

4. For measuring the time to cycle from school to home, a useful simple model would keep: 1

(a) the make of the tyres **(b)** distance and average speed
(c) the colour of the cycle **(d)** the school's name

5. The three main aims when approaching a numerical science problem are to understand the situation, identify the quantities that matter and then: 1

(a) use relationships and check with an estimate
(b) skip units
(c) memorise answers
(d) guess the final value

6. Smoke is an example of a heterogeneous mixture of: 1

(a) a gas in a liquid **(b)** solid particles in a gas
(c) a liquid in a solid **(d)** two gases

7. Particles in a true solution do NOT scatter light because: 1

(a) they are charged **(b)** they are coloured
(c) they are too large **(d)** they are too small

8. Which pair is a mixture of immiscible liquids? 1

(a) alcohol and water **(b)** oil and water
(c) acetone and water **(d)** vinegar and water

9. The pure liquid obtained by condensing vapour in distillation is called the: 1

(a) precipitate **(b)** filtrate **(c)** residue **(d)** distillate

10. Which of the following is a homogeneous mixture? 1

(a) sand in water **(b)** soda water (CO_2 in water)

(c) smoke in air **(d)** chalk powder in water

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why are weather forecasts usually reliable for a few days but not for much longer? 2

12. Why is estimation a useful scientific skill? 2

13. How would you separate a mixture of mustard oil and water? 2

14. Why can camphor be separated from sand by sublimation? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. How does a mask designed during a pandemic combine different branches of science? 3

16. State three properties of colloids. Give two examples. 3

17. Why does the textbook say that an equation is a ‘compact statement about how certain things are related’? Give an example. 3

OR

Explain why science values careful reasoning perhaps more than accurate calculations. 3

18. Why does a torch beam passing through a glass of milk-water mixture look visible while it is invisible through a glass of salt solution? 3

OR

What are the differences between evaporation and crystallization? State which gives pure crystals. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Meghnad Saha studied stars by treating the matter in a star as a hot gas. 5

(i) What did he ignore?

(ii) What did he focus on?

(iii) Why is this called a model?

(iv) What does this show about how science often begins?

OR

(i) How do laws, theories and models help predict? 5

(ii) Illustrate with the example of the weather.

(iii) State two limits of such predictions.

20. **Case study:** Oil spilled into water forms two layers while alcohol mixes completely with water. 5

(i) Classify the two mixtures.

(ii) How would you separate the first one?

(iii) How would you separate alcohol and water?

(iv) Why can the second not be done by a separating funnel?

OR

(i) What is an alloy? 5

(ii) How is an alloy prepared?

(iii) Why do we use alloys rather than pure metals?

(iv) Name the constituents of brass.

END OF QUESTION PAPER Total: 40 Marks